Lab - Azure SQL Database
To Begin with the Lab

Summary

This lab demonstrates how to create an Azure SQL Database using the Azure Portal. You will create a SQL Server, configure authentication and networking, create a database, connect to it using a SQL client or Azure Query Editor, verify the database is working, and finally delete the resources to avoid unnecessary charges.

Understanding

Azure SQL Database is a fully managed relational database service (Database as a Service - DBaaS) provided by Microsoft Azure. It is based on the Microsoft SQL Server database engine but eliminates the need to manage hardware, operating systems, backups, patching, and high availability. Azure automatically handles updates, security, backups, and scaling, allowing developers to focus on building applications instead of managing infrastructure.

Example:

Suppose you are developing an employee management application. Instead of installing Microsoft SQL Server on a virtual machine, you can create an Azure SQL Database. Your application connects to the database using a connection string, while Azure manages backups, security, and performance automatically.

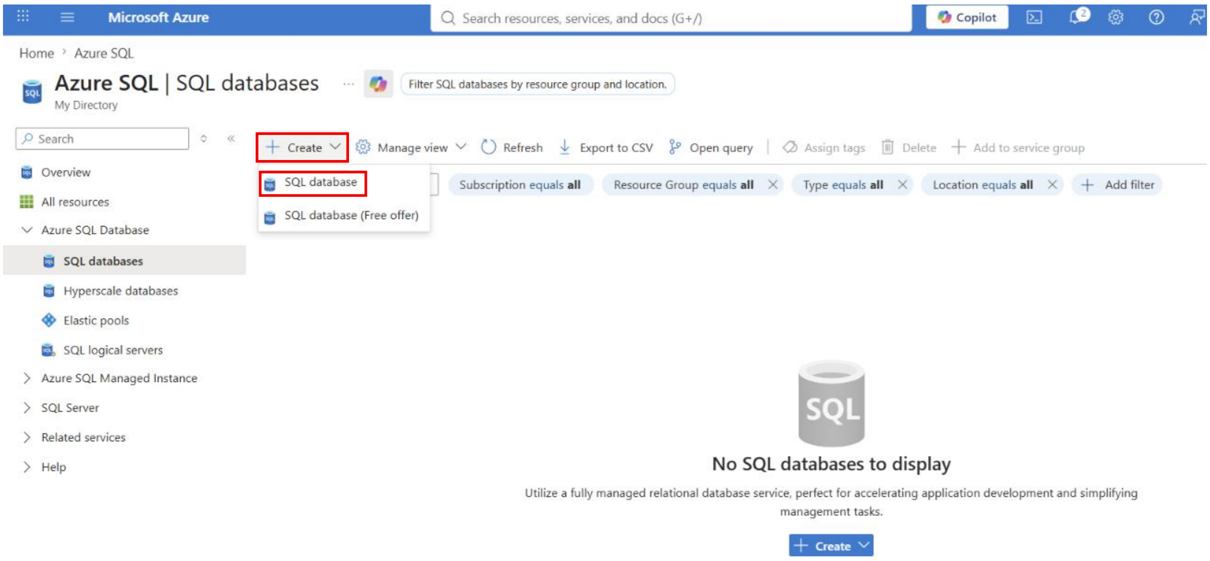
Lab steps

Step 1: Sign in to Azure Portal

- Open the Azure Portal.
- Sign in using your Azure account.

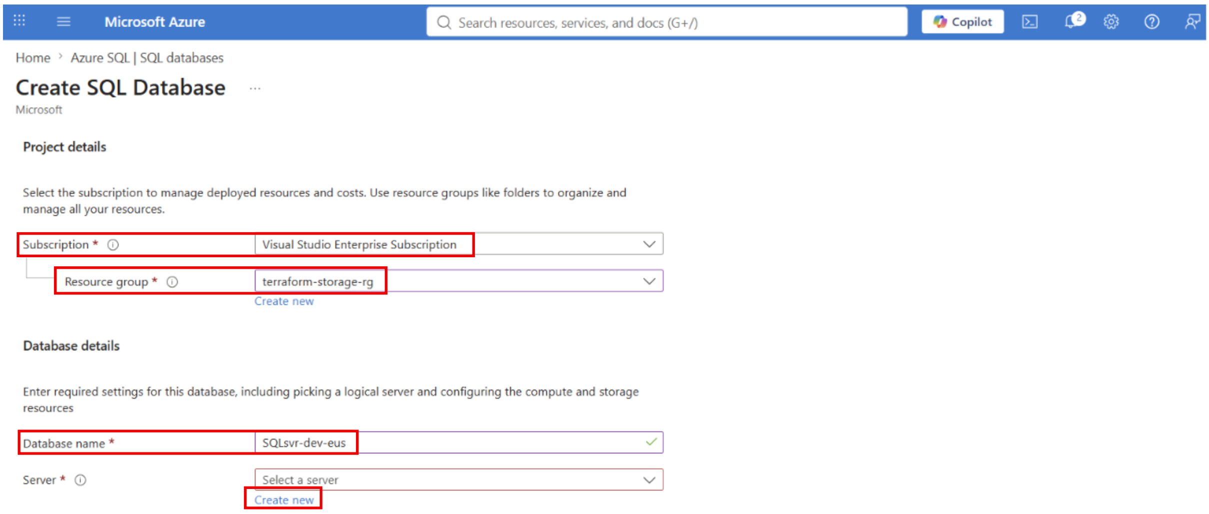
Step 2: Open Azure SQL

- In the search bar, search for **SQL Databases**.
- Open the **SQL Databases** service.
- Click **Create**.
- Select **SQL Database**.



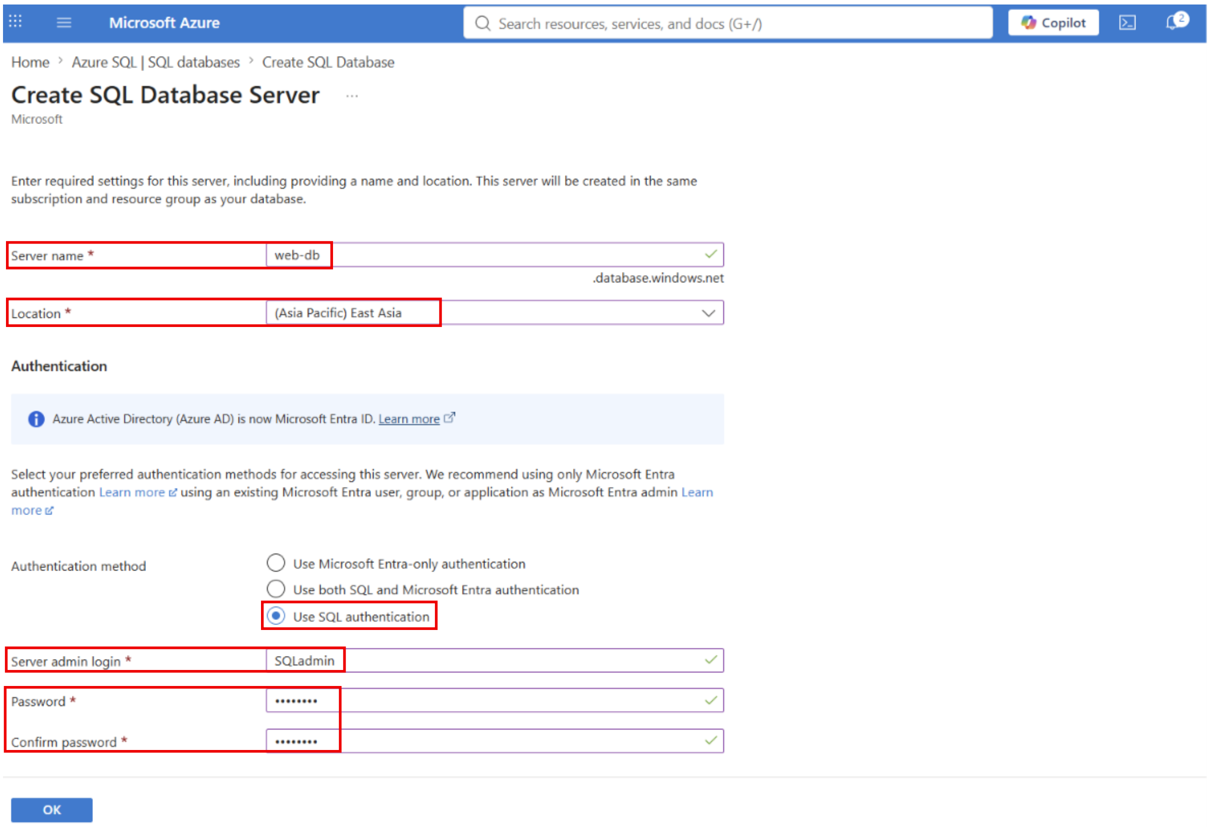
Step 3: Configure Basic Details

- Choose your Azure Subscription.
- Create or select a Resource Group.
- Enter a Database name.
- Create a new SQL Server or select an existing one.



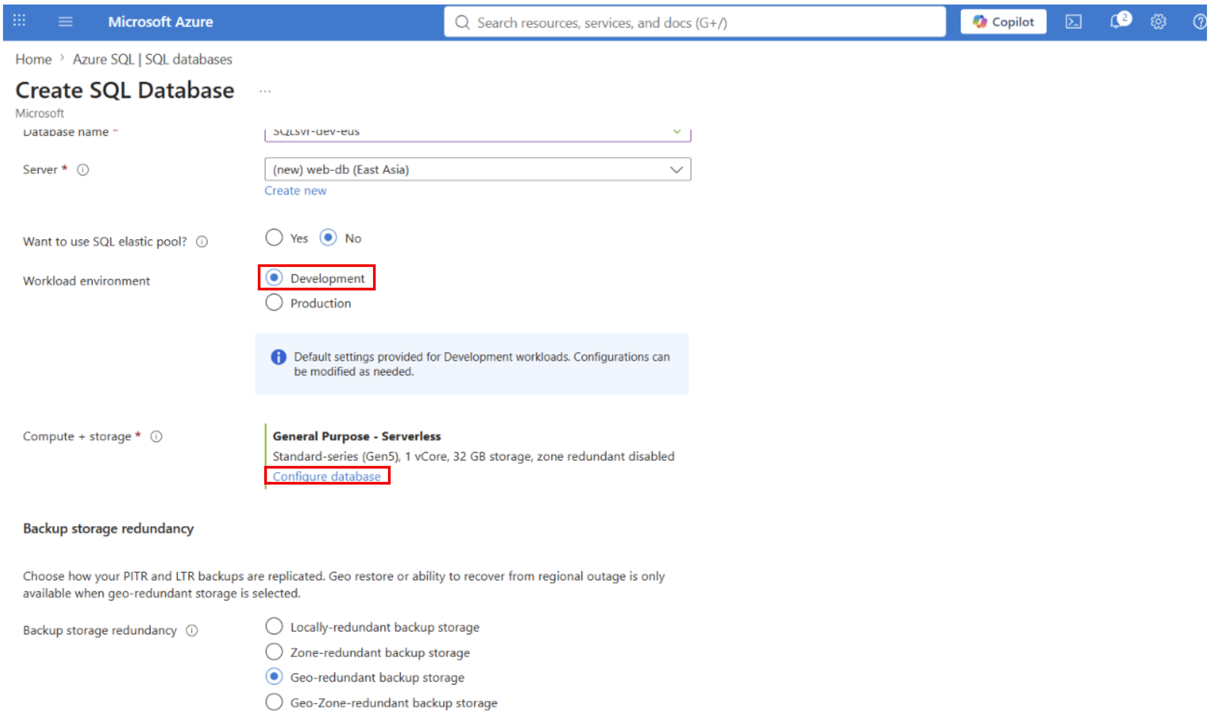
Step 4: Configure SQL Server

- Enter a **unique Server name**.
- Choose the **Region**.
- Select **SQL Authentication**.
- Enter the **Administrator username**.
- Enter and confirm the **Administrator password**.
- Click **OK** or **Review** to continue.

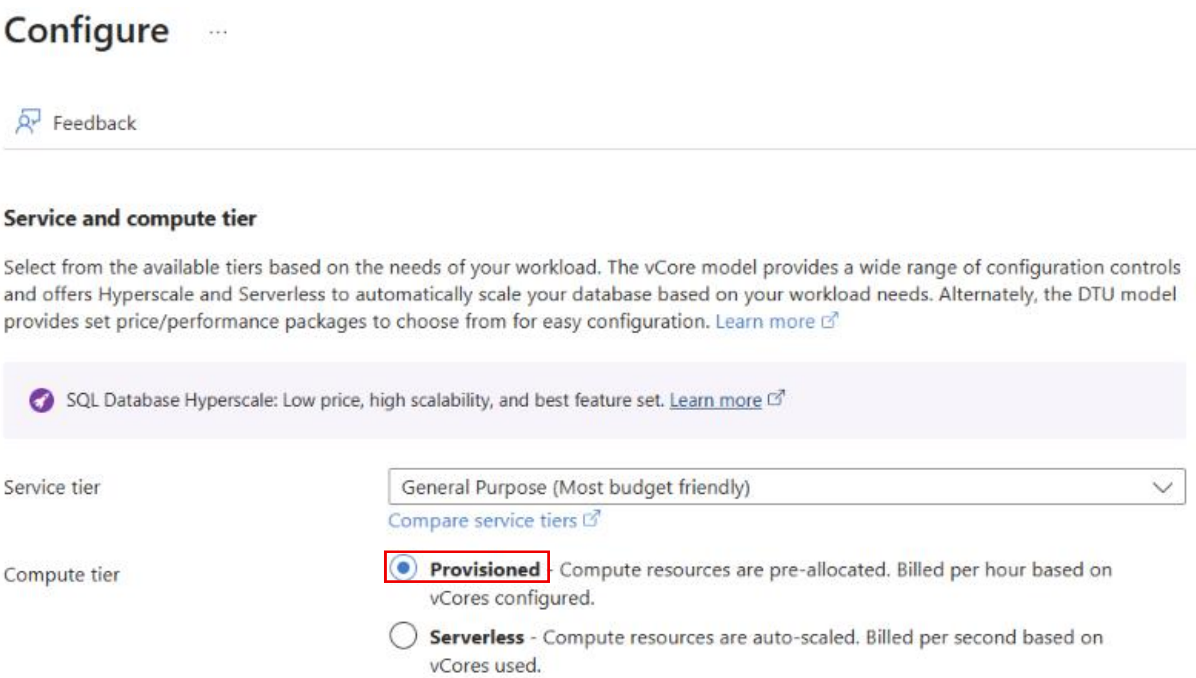


Step 5: Configure Compute and Storage

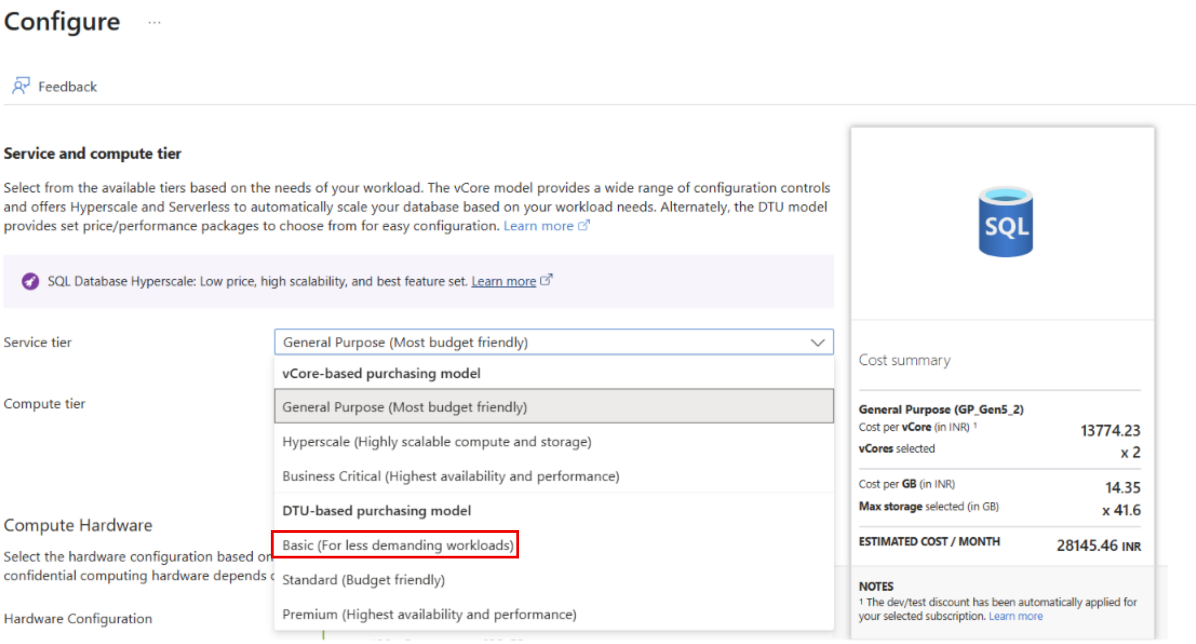
- Select the desired Workload type **development**.
- Click on the **Configure database.**



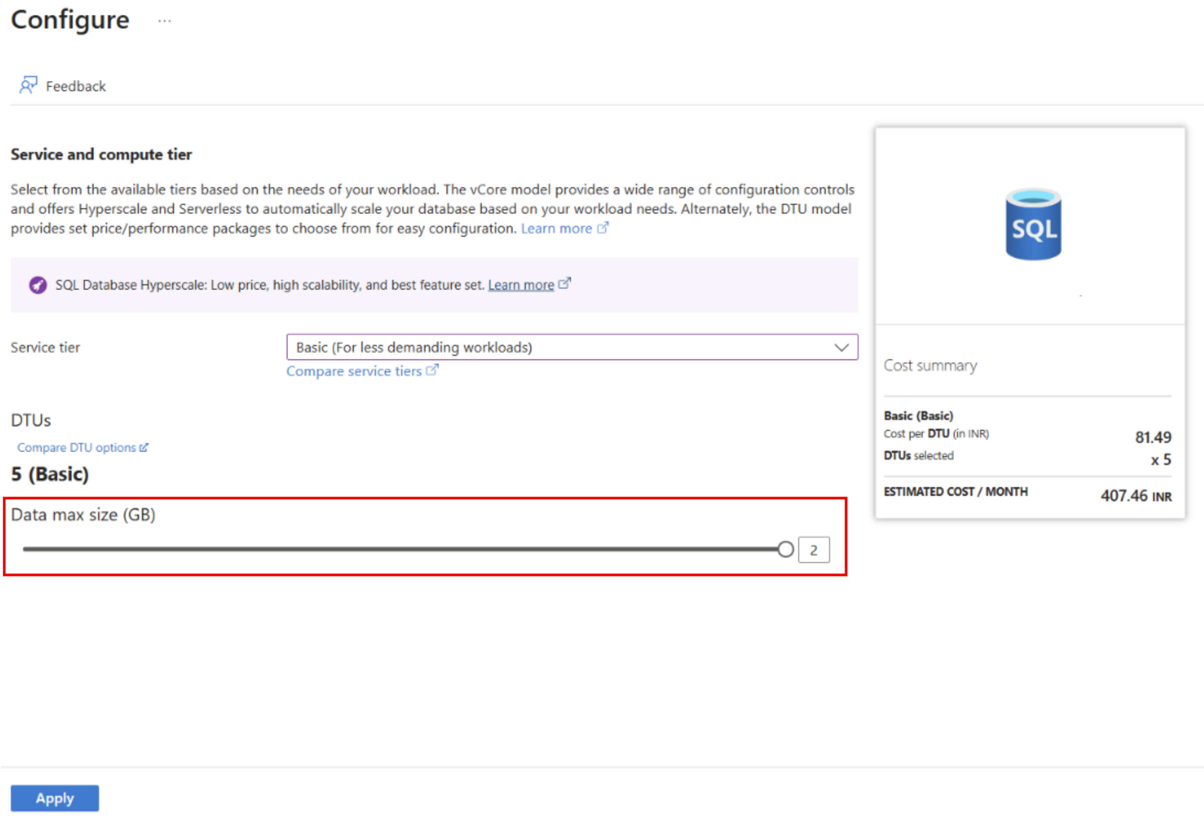
- Choose the Compute tier **provisioned**.



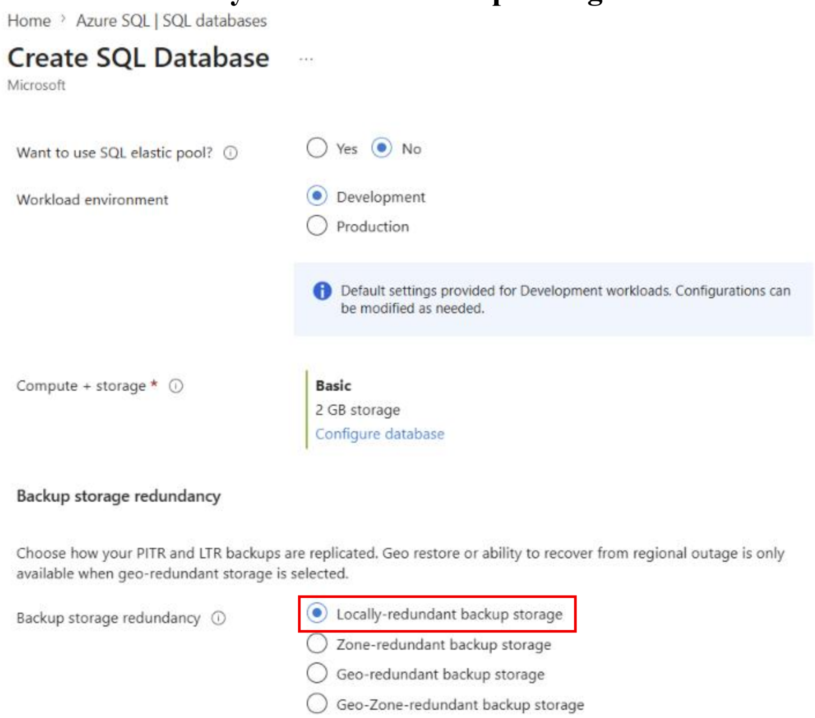
- Choose Service tier **Basic (For less demanding workload).**



- Select the required **Storage size**.
- Click on the **Apply**.



- Choose **Locally-redundant backup storage**

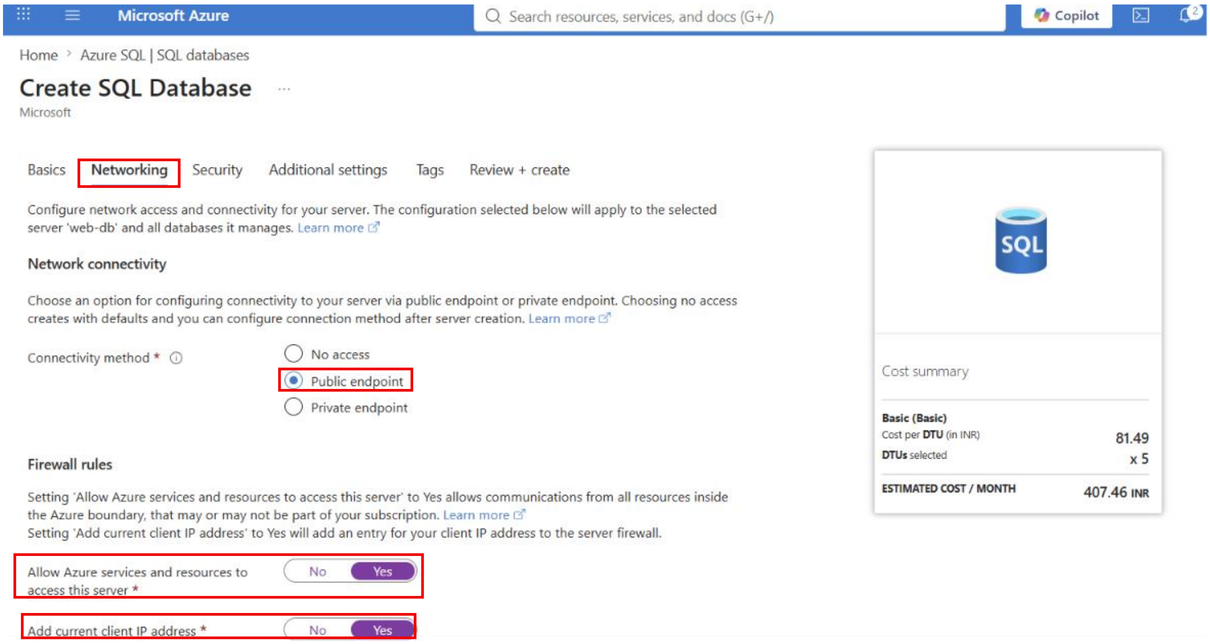


- Keep the default settings if performing the lab.

Step 6: Configure Networking

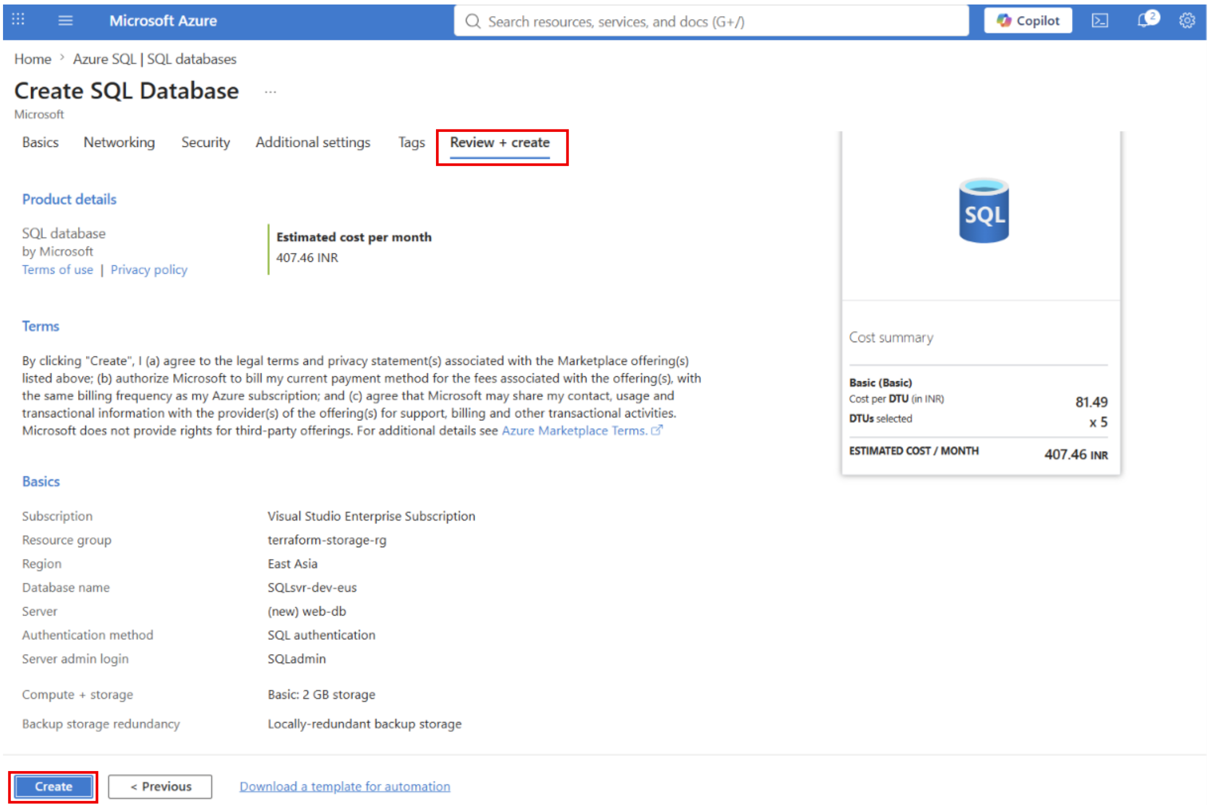
- Choose the Connectivity method **Public endpoint**.
- Enable **Allow Azure services and resources to access this server** if required.
- **Add your client IP address** to the firewall rules.

- Leave other networking settings as default unless specified.



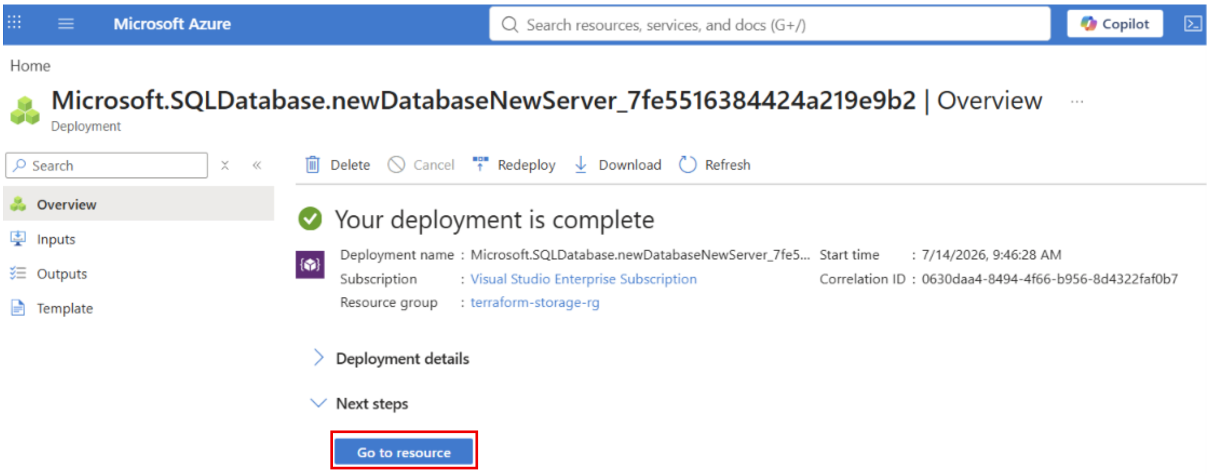
Step 7: Review and Create

- Review all the configuration details.
- Click **Review + Create**.
- Click **Create** to start deployment.



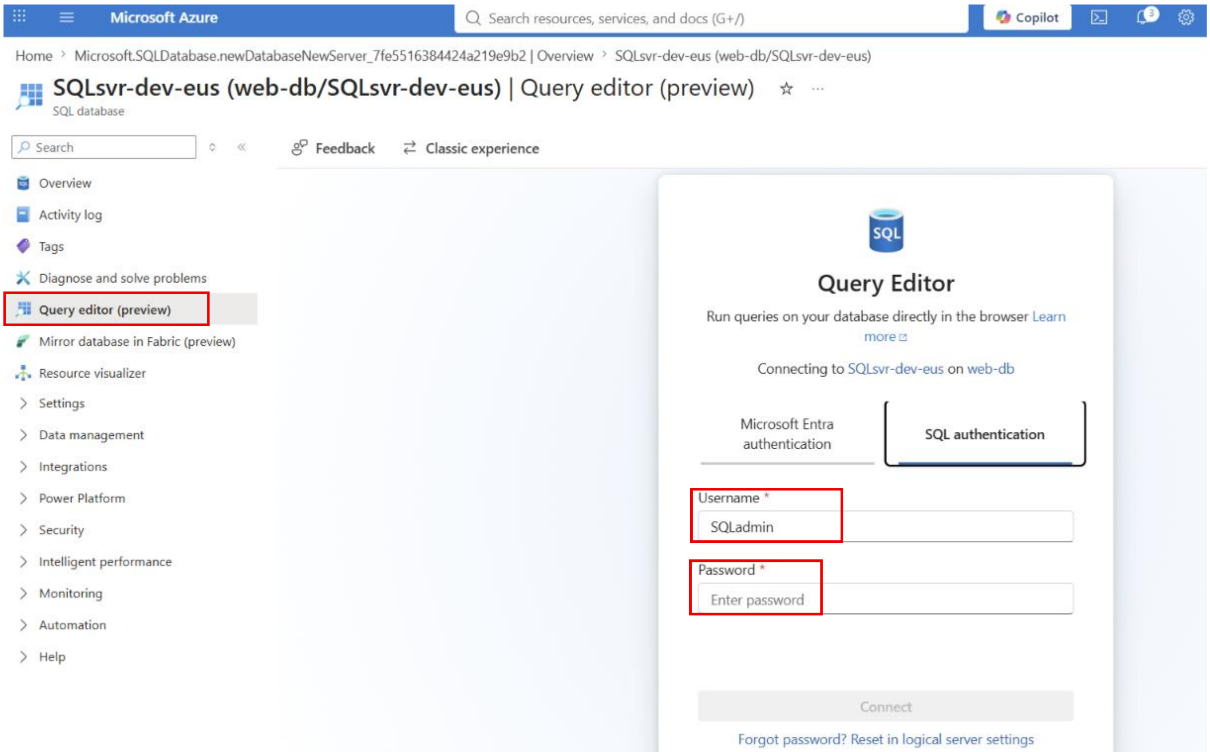
Step 8: Wait for Deployment

- Wait until the SQL Database deployment completes.
- Click **Go to Resource** after deployment is successful.



Step 9: Connect to the Database

- Open the SQL Database Overview page.
- Copy the Server name if required.
- Connect using Azure Query Editor
- Sign in using the SQL Server Administrator credentials.



Step 10: Verify the Database

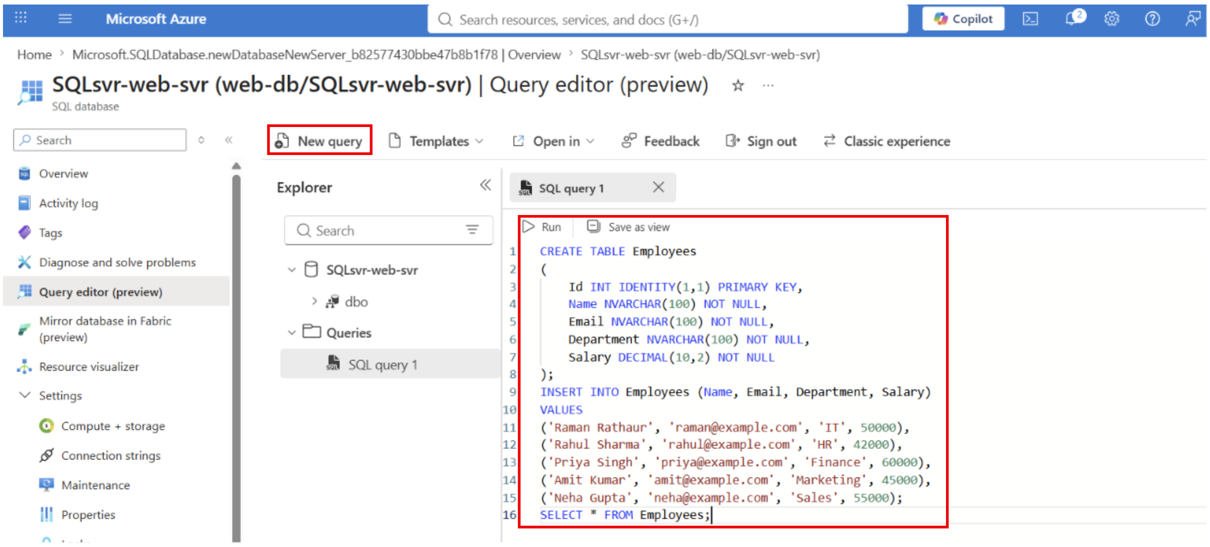
- Open the Query Editor.
- Create a sample table if required.
- Insert a few sample records.

```
CREATE TABLE Employees
(
   Id INT IDENTITY(1,1) PRIMARY KEY,
   Name NVARCHAR(100) NOT NULL,
   Email NVARCHAR(100) NOT NULL,
   Department NVARCHAR(100) NOT NULL,
```

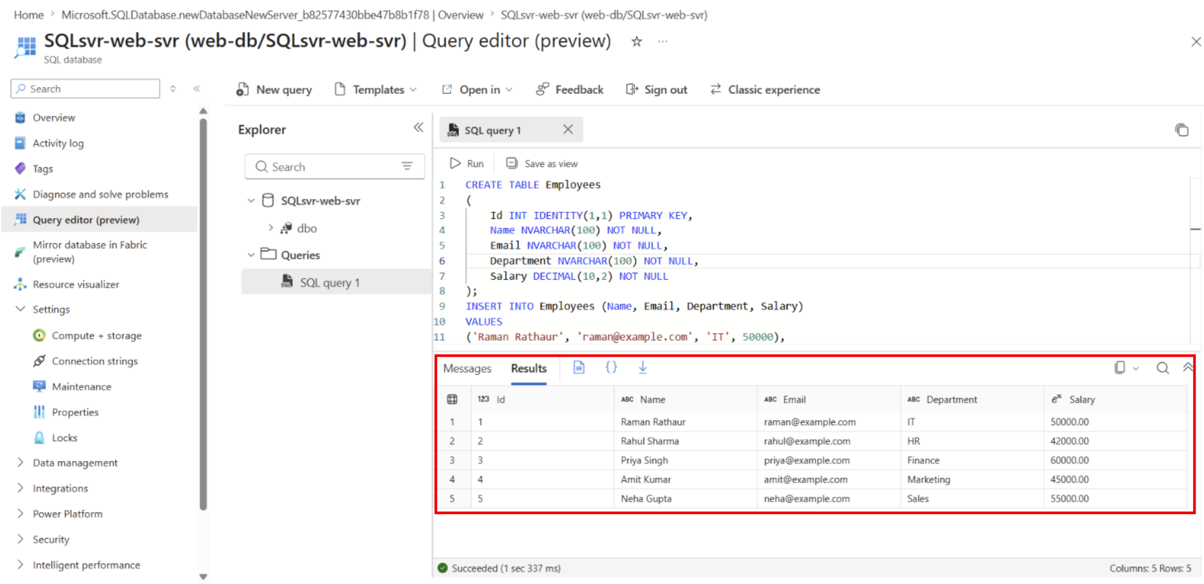
```sql
    Salary DECIMAL(10,2) NOT NULL
);
INSERT INTO Employees (Name, Email, Department, Salary)
VALUES
('Raman Rathaur', 'raman@example.com', 'IT', 50000),
('Rahul Sharma', 'rahul@example.com', 'HR', 42000),
('Priya Singh', 'priya@example.com', 'Finance', 60000),
('Amit Kumar', 'amit@example.com', 'Marketing', 45000),
('Neha Gupta', 'neha@example.com', 'Sales', 55000);
SELECT * FROM Employees;
```

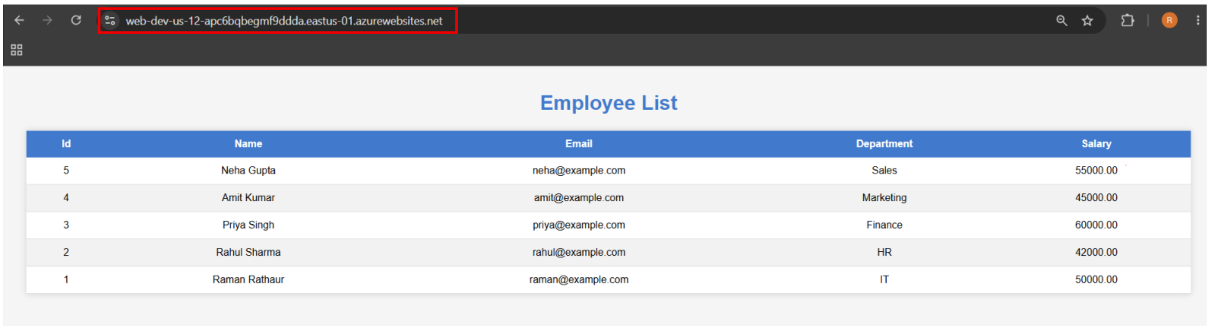
3. View All Records

- Click on the Run



- Run a **SELECT** query to verify the data.
- SELECT * FROM Employee;





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Employee List

Id	Name	Email	Department	Salary
5	Neha Gupta	neha@example.com	Sales	55000.00
4	Amit Kumar	amit@example.com	Marketing	45000.00
3	Priya Singh	priya@example.com	Finance	60000.00
2	Rahul Sharma	rahul@example.com	HR	42000.00
1	Raman Rathaur	raman@example.com	IT	50000.00